

Outperformed by AI: Interacting with Superhuman AI Changes the Way We Perceive Ourselves

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ABSTRACT

With AI increasingly exceeding human performance in previously quintessential human domains, the question posed in this study is how interacting with ‘superhuman’ AI will affect how we define 1) ourselves as human beings, and 2) our relationship towards the technology. To address this question, we conducted two playful quasi-experimental studies ($N_{\text{total}} = 349$). In both experiments, participants interacted with an AI system, and were, or were not, outperformed by it. Results consistently show that when AI outperformed the participants, they felt inferior, experienced less agency (control over the situation), and had a less positive attitude towards the AI system. This suggests that interaction with intelligent machines may well redefine our perceptions of what it means to be human in the long run and determine how we act towards this technology.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in collaborative and social computing*; **Empirical studies in HCI**; • **Computing methodologies** → **Artificial intelligence**; Philosophical/theoretical foundations of artificial intelligence; • **Applied computing** → **Law, social and behavioral sciences**;

KEYWORDS

Superhuman AI, Human-AI Interaction, Self-perception, Sense of Agency, Self-worth, AI Attitudes

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1 INTRODUCTION

In recent years, we have seen a number of mediagenic landmark AI events: Deep Blue vs Kasparov, AlphaGo vs Lee Sedol, Watson wins Jeopardy!, Siri, autonomous cars, and most recently ChatGPT. All provided instances in which the world took notice of AI’s ability to perform tasks that are commonly associated with humans [11].

Whether or not these events truly represent revolutionary technological steps remains a point of discussion. The fact, however, is that AI will become increasingly superior relative to human performance in at least some tasks that were previously the exclusive domain of humans [23]. This phenomenon is called superhuman performance [4, 19, 32, 45], and if the current acceleration of the technology, both in terms of capabilities and the range of areas in which it is applied, continues into the near future, this may pose questions of a more existential nature. One such question is how superhuman AI impacts how we define ourselves as human beings.

It has been acknowledged that superhuman AI has the potential to challenge people’s beliefs about the defining features of humankind [7, 17, 33, 46, 48]. For instance, research has shown that threats to our identity, including those by machines, can undermine motivations to act in domains that are related to the threat [7]. This raises concerns about the potential effects of advanced AI undermining for instance human autonomy [2, 22]. Thus, as superhuman AI technologies become more ubiquitous in daily life, it becomes crucial to understand how being confronted with intelligent technology outperforming us in tasks that used to be quintessentially human, will impact perceptions of ourselves as humans, as well as of the technology and our relationship to it [25]. Our research aims to answer these questions. Specifically, we address how interacting with AI outperforming the user affects a user’s perceptions of self-worth and sense of agency, and evaluation of the AI.

Although there has been limited research on this issue, the broader theme of how interacting with superior AI affects our perceptions of humanness has recently enjoyed an uptick in scholarly attention. First, research has revealed that people do often feel threatened by AI crossing the threshold of human-level capabilities and appearance [7, 17, 38, 42, 46]. A number of recent studies have shown that when faced with advanced AI in previously uniquely human domains such as cognition or the creative arts, people employ defensive mental strategies in order to maintain a view in which humans remain superior to machines [7, 33, 38]. Furthermore, a recent study investigated how altering the performance of AI in AI-human teams determines self-confidence and confidence in the AI [9]. Research of this kind has led some authors to conclude that advances in AI “pose a profound ontological threat” to humans [33].

Recent research also hints advances in AI are not only a threat to humans, as the advent of superhuman AI may have led to improved human decision-making [41]. What we do not know is how interacting with AI that outperforms us in a human domain task impacts how we think of ourselves and the AI. That is the purpose of the current investigation. Although these are potentially big questions, very little research has been done in this domain. Therefore, the current research presents a first, modest, and playful exploration

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into these questions, meant as a stepping stone for future work. We conduct two preregistered experiments in which participants are challenged to defeat an AI system capable of facial recognition (Experiment 1) or generating art (Experiment 2).

We find that it matters whether participants ‘win’ or ‘lose’ in terms of their feelings of relative self-worth, sense of agency, and attitudes towards AI. If these results can be extrapolated to the longer-term and other types of AI, this may mean that humankind’s self-image may suffer from being increasingly surpassed by intelligent machines.

2 BACKGROUND: THE IMPACT OF SUPERHUMAN AI

There are several conceivable areas in which AI exceeding human performance may impact users’ (self-) perceptions when interacting with the AI. As a first exploration, we focus on four such potential areas: a user may have (1) lower self-worth, (2) a decreased sense of agency, and a lower evaluation of (3) the interaction and (4) the AI itself.

2.1 Self-worth

Superhuman AI may impact a subjective feeling of inferiority, or self-worth relative to the AI in two ways: people may see machines’ (increasing) capability of performing human tasks as a threat to either human and personal uniqueness [38, 46], or it may affect personal self-esteem through negative feedback [40].

First, a threat to human uniqueness centers around the uneasiness when people see AI performing tasks that were previously uniquely reserved for humans. Oh et al. [34] report that although before the AlphaGo vs Lee Sedol game, viewers were optimistic about Lee’s chances to win, after his defeat, they were struck by feelings of helplessness, depression, and a sense of human frailty. Several recent experiments have shown that when participants feel threatened by AI on one evaluative dimension, for instance cognitive, they switch to other dimensions for evaluation, such as emotional responsiveness. It is believed participants switch domains in order to maintain an image of humans as distinct and superior to machines: *they* may be better than us in chess and go, but *we* are still superior in having and interpreting emotions [7, 38]. In addition, the degree to which users regard themselves as unique human beings [30], or perceive they are being treated as such by AI [24], has been found to be related to AI acceptance and satisfaction.

A second pathway to reduced feelings of relative self-worth may be through negative feedback. In psychology, receiving negative performance feedback (i.e., informing a person of their inadequate performance) has been shown to affect a person’s self-worth and self-esteem [3, 27], and the same is true when people receive negative feedback from a computer [44]. A recent study provides additional indications that AI performance relative to human performance can affect user self-confidence [10]. Using scenarios, it concludes that when users receive positive feedback after rejecting an AI suggestion, they start to look down on AI whenever its performance deteriorates, due to increased self-confidence. Conversely, the users lose self-confidence when they receive negative feedback after accepting a suggestion.

In sum, as people perceive AI as outperforming humans, specifically in tasks that were previously strictly a human prerogative, the outperforming poses an ontological threat to human distinctiveness. Moreover, negative feedback – presumably including feedback on losing to an AI in a challenge – likely affects the feelings of self-worth. Therefore, we expect that ***if participants are outperformed by the AI, their sense of self-worth relative to the AI will be lower, compared to participants who are not outperformed (H1).***

2.2 Sense of Agency

One core aspect that defines our humanity is our sense of agency—the perceived ability to control our actions to achieve desired outcomes. The human species has evolved the capability of making choices and to act on these choices in ways that allow it to adapt to its ever-changing environment [1, 28, 31]. A sense of agency is grounded in the belief that one is capable of outlining and executing actions to attain certain goals [1]. In other words, if people believe they can do something (i.e., have a high ‘self-efficacy’), their sense of agency is high. Conversely, if one is less convinced of one’s capabilities, the sense of agency would be lower.

Authors in Human-Computer Interaction research have suggested that sense of agency is a key factor in how people experience interactions with technology (e.g., [2, 13, 29]). Additionally, neurocognitive experiments using motor feedback have found that if the outcome of the task is unexpected to the participant, this leads to a loss of sense of agency [14], or agency being attributed to other sources than the self [43]. Importantly, negative feedback on performance impairs self-efficacy [5]. In our studies, the idea is that ***if participants are outperformed by the AI, their sense of agency will be lower, compared to participants who are not outperformed (H2).***

2.3 Evaluations

Recent research has seen increased interest in AI attitudes (cf. [20, 26, 47]). Similar to the reasoning above, threat perceptions may play an important role in how people evaluate the interaction with AI and the attitudes they develop towards it. Stein et al. [42] developed the Model of Autonomous Technology Threat, in which they propose and empirically demonstrate that perceived threat to both situational control and to human uniqueness predicts stronger aversion to intelligent technology. Likewise, Złotowski et al. [48] find that perceived threats to human uniqueness by people watching a video of autonomous robots evoked negative attitudes towards robots in general. Finally, Bergdahl et al. [2] (Study 1) find that using new technologies can make people feel insecure about their abilities, and that feeling incompetent leads to more negative attitudes toward the technology. Thus, we expect that ***if participants are outperformed by the AI, their evaluation of the interaction will be more negative, compared to participants who are not outperformed (H3)*** and ***if participants are defeated by the AI, their attitudes towards the AI will be more negative, compared to participants who are not outperformed (H4).***

3 METHOD

To test our hypotheses, we designed two experiments that allowed ‘ordinary’ people to interact with an AI system. The idea was to challenge participants to a game against AI in a domain that could be perceived as - until recently - requiring exclusively human capabilities. We chose two such domains: facial recognition, and producing art. The reasoning was that when people are defeated by a system in a classic human task, this would trigger a reevaluation of their self-perception as human and of the AI. To ensure the game and its outcomes were as realistic as possible, we let the outcome — and by extension the participant’s skill in the game — determine whether the participants was grouped in the ‘win’ or ‘lose’ group.

3.1 Design

We conducted two separate 1-factorial 2-group between-subjects quasi-experiments to investigate the effects of interacting with (and losing to) an AI system that is capable of performing tasks previously reserved for humans: In experiment 1, the AI system is able to classify human faces from other objects. In experiment 2, it can generate believable ‘oil paintings’. Both experiments were administered online, preregistered, and approved by the ethical committee of the authors’ institution.¹

In both experiments, an AI system was presented as an opponent in a playful, game-like setting. Participants were divided into two groups based on whether they won or lost against the AI (thus producing the factor: ‘win’ or ‘lose’) in both studies.

3.2 Samples

The G*Power software [16] was used to conduct a power analysis; indicating that a sample size of $N = 128$ would be sufficient to obtain .80 power to detect a medium effect size of $f = .25$ at the standard .05 alpha error probability. We recruited a sample of >18 y.o. UK residents, whose 1st language is English from the online participant recruitment platform Prolific (<https://www.prolific.co>). Participants were paid €1,10-1,35 for 7-10 minutes of participation. Participants failing an instructed response test (attention check) and those who completed the questionnaire in under 2 mins. were excluded from the analyses and replaced. Only fully completed questionnaires were accepted, resulting in a sample of $N = 210$ for Experiment 1 (Age $M = 41.61$; $SD = 13.67$; 57.6% Female; 20% graduate degree or higher; 12.4% educational/professional background in computer science, IT, AI or related disciplines), and $N = 139$ for Experiment 2 (Age $M = 40.68$, $SD = 14.09$; 52.5% Female; 15.1% graduate degree or higher; 14.4% background in computer science and related disciplines; 5% background in arts and related disciplines).

3.3 Procedure

After filling out informed consent, in Experiment 1, participants received instructions telling them: “You will now interact with an Artificial Intelligence (AI).” After a short explanation as to the nature of the AI, they were instructed to: “try to fool the AI into misidentifying you as non-human. You will take a ‘selfie’ (that is, a self-portrait) with your webcam, and show this selfie to the AI. You can use any trick you like to fool the AI, but you should be visible in the image. If

the AI identifies you as anything other than ‘human’, you have won! If, however, the AI succeeds in identifying you as human, the AI has won.” Participants then would interact with the AI system/website. The system first greeted users, asked for the permission to access the user’s camera, and subsequently, showed a mirrored version of their webcam input and a button to take the selfie. Upon taking a selfie, participants had the option to re-take the picture (as often as desired) or to have it analyzed by the AI. After having the image analyzed, the AI’s verdict was presented in the form of a short text. The message informed them that they are a human, or that they are something else, indicating the object or animal the AI mistook them for (see Figure 1). In addition, a text was displayed that asked the participant to report the result in our questionnaire. They then completed our questionnaire.

In Experiment 2, we informed participants that they would “play a game against an Artificial Intelligence (AI) system.” After a short explanation about the AI, they were informed that they would see a painting and that they needed to identify the painting as either human or AI-created, explaining: “If you decide correctly, you win! If you are mistaken, you lose, and the AI system wins.” After viewing the image, participants indicated their answer. Depending on their choice, they saw a screen telling them either “You lose! The painting was made by an AI system called Midjourney. It seems the AI had you fooled!”, or “You win! The painting was made by an AI system called Midjourney. It seems the AI could not fool you!”. They then answered questions on the dependent variables and control variables.

3.4 Materials

The system used in Experiment 1 took the form of an interactive website. At <https://misidentify.me>, one can find a public version that closely resembles the experiment website. The system is called ‘Misidentify.me’ and it is capable of detecting humans as well as identifying a range of objects and animals. We previously presented it at a science film festival and describe the system in an accompanying paper [39].

In Experiment 2, we used an image generated by the generative AI program Midjourney (www.midjourney.com), using the prompt “a day at the beach oilpainting --ar 4:3”² All participants were shown the same image (see Figure 2), which was entitled ‘A Day at the Beach’. The image was selected from several in a pilot study where 50% of the participants judged it as ‘human-made’.

3.5 Measures

Both experiments were administered online, using Qualtrics questionnaires, with the scales and questions described below. For all scales, scores were averaged so higher scores represent higher levels of the measured variable.

Self-worth relative to the AI was measured using a 7-point semantic differential scale with 7 items about feelings of relative self-worth; e.g. “I am smarter than the AI-The AI is smarter than me” (Experiment 1: $M = 3.87$, $SD = 1.64$, $\alpha = .95$; Experiment 2: $M = 3.64$, $SD = 1.30$, $\alpha = .93$).

Sense of Agency was measured using a 7-point semantic differential scale with 4 items. The items probed the degree of perceived

¹Preregistrations and all data, analysis protocols and study materials are available at https://osf.io/y25ec/?view_only=7dd315f6e749481d84e1060df3155c7c.

²the ‘-- ar 4:3’ specifies the aspect ratio.

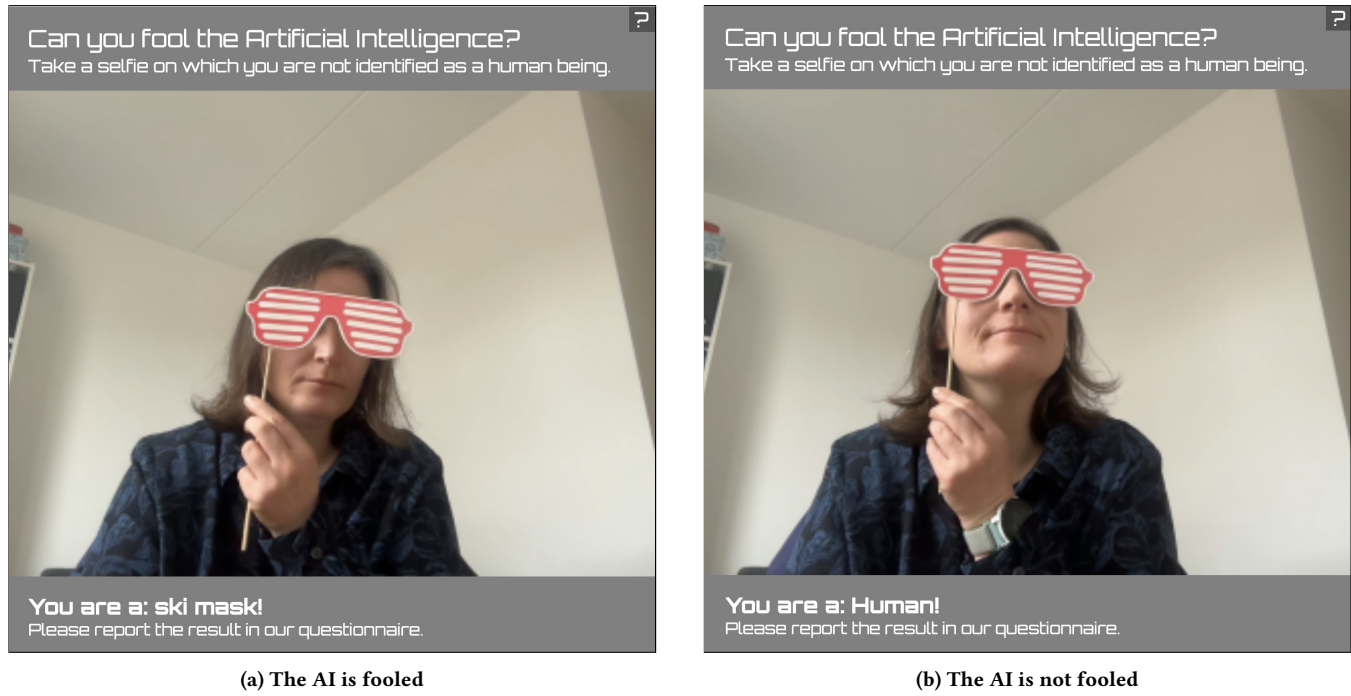


Figure 1: Misidentify.me: The result-screen

control over the situation; e.g., “I determined the outcome-The AI determined the outcome” (Experiment 1: $M = 4.03$, $SD = 1.64$, $\alpha = .84$; Experiment 2: $M = 3.91$, $SD = 1.47$, $\alpha = .80$).

Interaction Evaluation was measured using a 7-point semantic differential scale with 11 items, including, e.g. “Pleasant-Unpleasant”,

“Fun-Not fun”; “Educational-Uninformative”, and “Entertaining-Dull” (Experiment 1: $M = 3.68$, $SD = 1.17$, $\alpha = .91$; Experiment 2: $M = 3.14$, $SD = 1.01$, $\alpha = .90$).

Attitude towards the AI System participants interacted with was measured with a 7-point differential with 12 items; e.g. the system could be described as “Smart-Stupid” and “Good-Bad” (Experiment 1: $M = 3.34$, $SD = 1.05$, $\alpha = .91$; Experiment 2: $M = 3.10$, $SD = 1.00$, $\alpha = .92$).

General Attitude towards AI of this type being applied in real life was measured with a 7-point semantic differential scale with 11 items; e.g., they could indicate whether they find this type of AI “Good-Bad”; “Positive-Negative”; “Unalarming-Scary” (Experiment 1: $M = 3.20$, $SD = 1.14$, $\alpha = .95$; Experiment 2: $M = 3.49$, $SD = 1.07$, $\alpha = .92$).

Demographics and Control Variables. We asked participants’ age, gender, education, and professional or educational background in AI, computer science, or related field. No differences were found between the groups. We therefore conclude that participants distributions across the groups are not affected by demographics or control variables.

Analysis Strategies. We conducted separate independent sample t-tests for agency, superiority, experience, and attitudes as the dependent variables, each with ‘outcome: win/lose’ as grouping variable. Due to the self-distribution of participants across conditions according to the outcome of the challenge, sample sizes were unequal between the groups. However, Levene’s tests show that in all dependent variables, variances were equal between the groups.



Figure 2: A Day at the Beach, by Midjourney (6th October 2022).

4 RESULTS

First, we determined how successfully participants defeated the AI. Interestingly, participants were unevenly distributed across the ‘win’ or ‘lose’ conditions, with a majority losing to the AI (65.7% in experiment 1 and 73.4% in experiment 2).

Next, we tested the hypotheses. Figure 3 provides a summary of all effects. Hypothesis 1 stated that the participants’ self-worth relative to the AI would be lower in case they were outperformed by the AI. In Experiment 1, this was indeed the case: participants who lost had lower relative self-worth ($M = 3.20$, $SD = 1.28$) than participants who won ($M = 5.13$, $SD = 1.16$). Interestingly, participants that were outperformed felt genuinely inferior to the AI, as suggested by the fact that their mean scores were below the midpoint of the inferiority-superiority scale (4). A t -test showed the difference was significant $t(208) = 10.718$, $p < .001$, $d = 1.558^3$. The same was found in Experiment 2: participants who lost ($M = 3.30$, $SD = 1.21$) scored significantly lower than participants who won ($M = 4.55$, $SD = 1.07$); $t(137) = 10.718$, $p < .001$, $d = 1.558$. Thus, Hypothesis 1 was accepted.

Hypothesis 2 stated that losing decreases participants’ sense of agency. This proved to be the case in both experiments. In Experiment 1, a t -test showed that the difference between participants who won ($M = 5.12$, $SD = 1.43$) and who lost ($M = 3.45$, $SD = 1.44$) was significant: $t(208) = 7.964$, $p < .001$, $d = 1.158$. Likewise, in Experiment 2, participants who lost had a diminished sense of agency ($M = 3.50$, $SD = 1.34$) compared to winning participants ($M = 5.03$, $SD = 1.21$), $t(137) = 6.082$, $p < .001$, $d = 1.167$. Again, the means showed that participants had a low sense of agency (below the scale midpoint) when outperformed, whereas the ‘winners’ had a sense of agency well above the midpoint. Hypothesis 2 was thus supported.

In Hypothesis 3, we expected participants to have a more negative evaluation of the interaction when they lost. However, this hypothesis was not supported in either of the two experiments. In Experiment 1, the evaluation when winning ($M = 3.50$, $SD = 1.04$) was almost the same when losing to the AI ($M = 3.77$, $SD = 1.22$), $t(208) = -1.600$, $p = .111$, $d = -.233$; in Experiment 2, we saw the same results ($M = 3.06$, $SD = 1.01$ vs $M = 3.17$, $SD = 1.01$), $t(137) = -.534$, $p = .594$, $d = -.10$.

Finally, Hypothesis 4 predicted that losing would result in more negative attitudes towards the AI. We evaluated this with respect to the specific AI system used in the experiment, and towards ‘this type of AI’ in general. In Experiment 1, participants had a more negative attitude towards both the specific AI they interacted with and this type of AI in general after they lost (Specific Attitude $M = 2.98$, $SD = .93$; General Attitude $M = 3.0$, $SD = 1.09$) compared to participants who won (Specific Attitude $M = 4.03$, $SD = .91$; General Attitude $M = 3.58$, $SD = 1.14$). Both were significant differences: $t(208) = 7.834$, $p < .001$, $d = 1.14$; and $t(208) = 3.591$, $p < .001$, $d = .05$. In Experiment 2, we found the same result for the specific AI ($M = 3.51$, $SD = 1.22$ vs $M = 2.95$, $SD = .87$), $t(137) = 2.979$, $p = .003$, $d = .57$, but not for the AI in general ($M = 3.75$, $SD = 1.19$ vs $M = 3.40$, $SD = 1.01$), $t(137) = .1713$, $p = .09$, $d = .33$. Therefore, we can only partially accept Hypothesis 4.

5 CONCLUSIONS AND DISCUSSION

The continuing improvement and increasing ubiquity of AI suggests that encounters in which humans are outperformed in areas that were previously a human domain may become more frequent. Our well-powered and preregistered research indicates that this may not leave the human in the interaction unaffected. In two playful studies, we consistently find that people who are outperformed by AI in a human task perceive themselves as being inferior compared to the AI, and have a diminished sense of agency. Furthermore, we find that their attitude towards the particular AI they interacted with is more negative. An insignificant result in one experiment leaves it uncertain if and how the attitude towards AI in general is affected by being outperformed. Furthermore, perhaps because of the ‘fun’ game-like setting, we found no evidence that the experience itself of interacting with AI is diminished by being outperformed.

The current research presents a first exploration into a near future in which frequently encountering superhuman AI may have effects on how humans perceive themselves and the technology. Although by no means all-encompassing in terms of human self-image measures, relative self-worth and agency are seen as important factors in determining how people define themselves and their actions. If diminished for longer duration, both can lead to increased passivity and depression [12, 35]. Importantly, the performance challenges in the current research did not involve inherently threatening skills. But as even these seemingly trivial skills produced the effects we found, we may wonder what the effect will be if and when AI starts to outperform us in more fundamental life areas, threatening our self-image in vital domains such as work, decision-making, and relationships.

Our research contributes to the literature on the effects of interactions with AI on attitudes in two ways. First, we confirm recent research findings that a perceived ontological threat emanating from interactions results in negative evaluations of the AI [2, 42, 48]. But we extend that research by showing that such threat perceptions also affect evaluations of ourselves in various ways. Likewise, our second study contributes to the increasing body of research in which participants are asked to judge whether a human or an AI/computer has created an artwork [8, 15, 18, 36, 37] and studies about humans’ ability to distinguish real and synthetic images [6] and videos (i.e., deepfakes) [21]. Whereas existing research has predominantly focused on whether humans can identify generated images and how the (perceived) identity of the author (human or AI) affects *perceptions of the artwork*, our study introduced a novel focus on how AI’s abilities can affect human *self-perception*. To the best of our knowledge, this dimension has not been discussed in such prior studies.

Our research has brought the question if and how interaction with current and future superior AI may affect humanity to the HCI community. One plus of our research may be that, while research into the effects of AI often uses scenario-based research designs, we were able to conduct research based on real human-AI interactions. Nevertheless, this work is only a first exploration of these questions, which may raise many more questions than it answers. The current study is limited in terms of the longevity of the effects it is testing: although we find consistent effects of being outperformed by AI on psychological states, this of course does not necessarily mean

³Cohen’s d

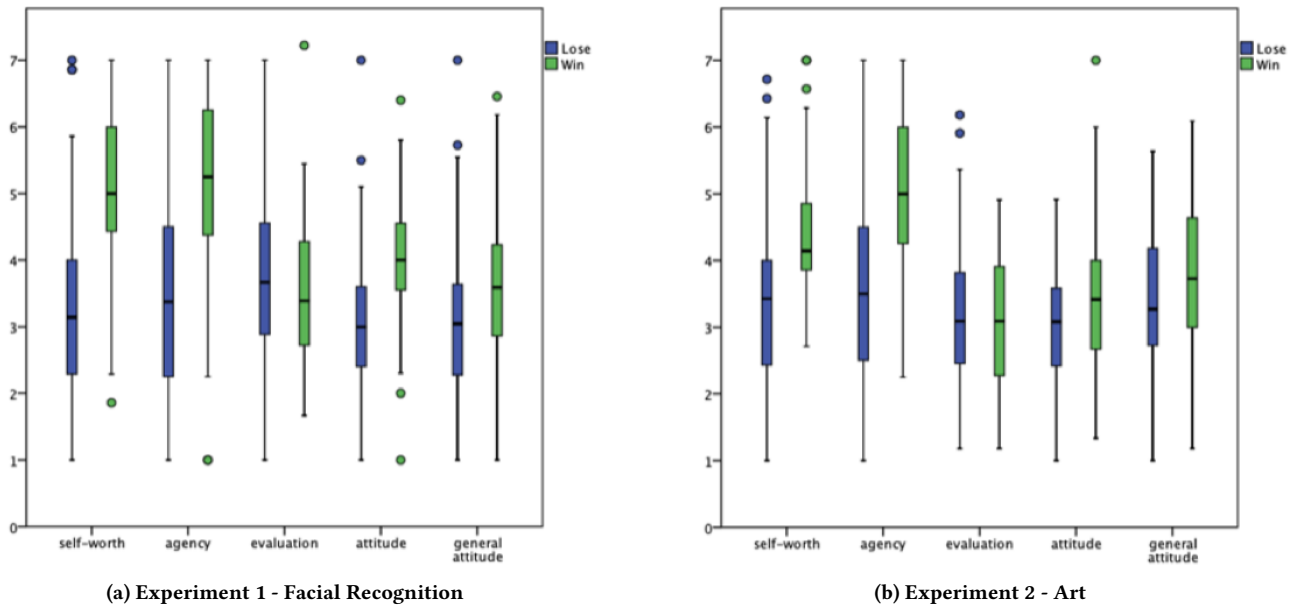


Figure 3: Effect of Winning or Losing to AI: Summary of Results

that we can expect that mankind will regress into chronic feelings of low self-esteem and lack of control. Due to its design, the current research does not allow this type of conclusion. Therefore, a question raised by the current work for future research is whether the short-term effects found in this study are non-trivial in case of longer-term exposure to superhuman AI. Similarly, follow-up work can investigate in how far similar effects are found when interacting with different types of AI systems (e.g., Large Language Models), and in settings where human and machine both need to carry out the *same* task. Finally, another question for future work is whether the effects found in the study, if true, are reflective of other effects on human qualities, such as acceptance of and reactance to AI, motivation, and ultimately, behavior.

A curb on any definitive conclusions concerns the quasi-experimental design of our studies. Any researchers triggered by the questions raised by our research should therefore focus on full experimental designs to ascertain the causal validity of our results, for instance by manipulating AI performance levels. Specifically, it would be interesting to confirm that our identified effects can indeed be attributed to the *intelligent nature* of the machine outperforming the participants and/or whether losing to *any* kind of machine has similar effects. For now however, this first attempt offers a kind of ‘proof of concept’, providing indications that it is not outside the realm of possibilities that if more and more previously quintessentially human talents and behavior will be performed at a higher level by machines, this may redefine our perceptions of what it means to be human in the long run.

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